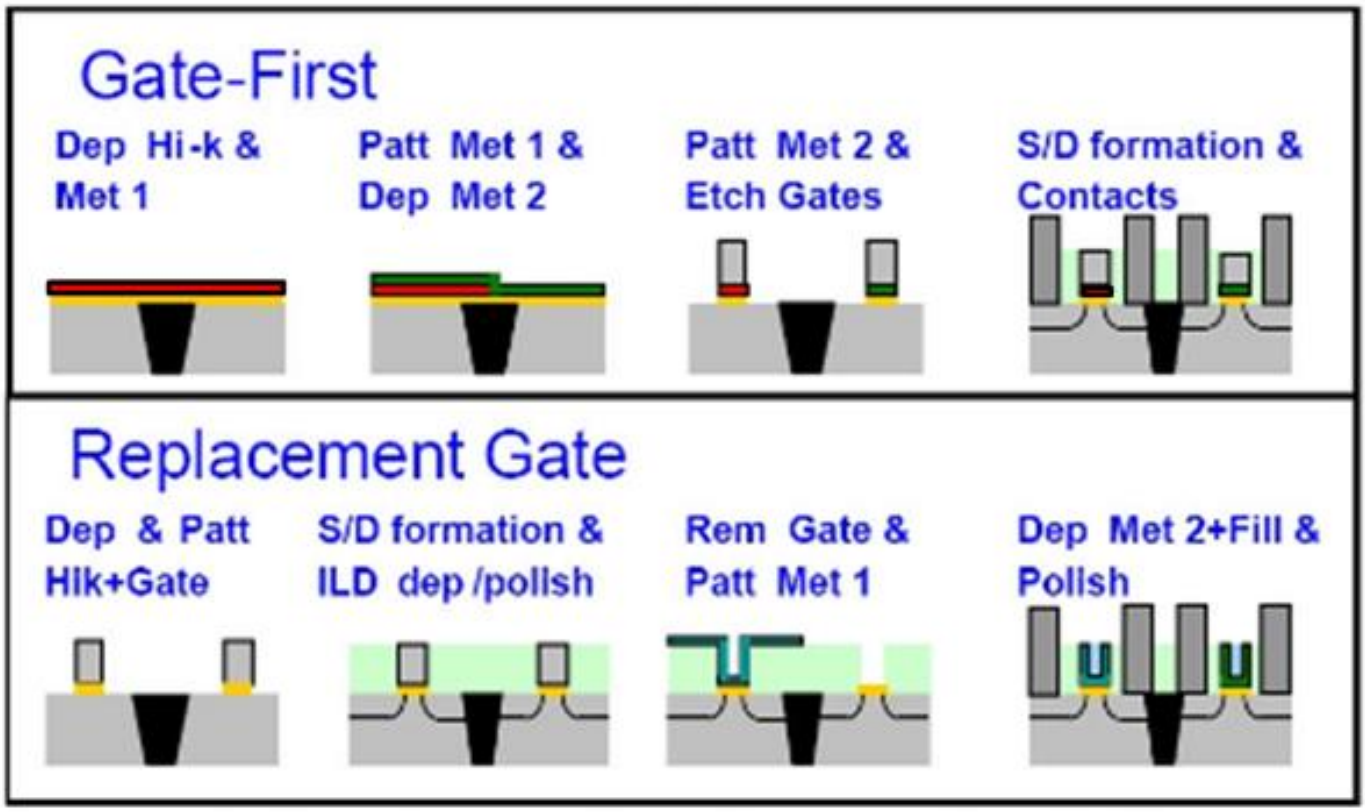
Assignment 4: Metal-First/Metal-Last



Learning outcomes

1. Difference between metal-first and metal-last (replacement gate) process.
2. Need of replacement metal gate due to thermal budget constraints
3. Importance of conformal deposition in structures with high aspect ratio.

Starting resources

1. Files:
    a. Assignment4_Gatefirst.zam
    b. Assignment4_GateLast.zam
    c. Inbuilt example of 14 nm logic process (14nmFinFET.zam).
2. Reference:
    a. https://sst.semiconductor-digest.com/2010/03/integrating-high-k/
    b. “High-K materials and Metal Gates for CMOS applications” by J. Robertson & R.M. Wallace. Section 1 & 2.

Questions

At the end of this assignment, we expect a report. Add images or graphs to support your conclusions and answers.

1. Please open Assignment4_Gatefirst.zam. Build the project. Open the viewer. Understand the structure. Look at the cross-section. Look at the process steps. Notice that the gate stack is deposited before the high-temperature anneal steps for the source and drain.
    a. The high-anneal step leads to metal diffusion. Plot the Al metal diffusion in the structures. What is the concentration in the active region?
    b. The device stops working if the metal contamination exceeds 10^{12} cm-3. Go to the anneal recipe and reduce the time of post-implant anneal. What is the maximum time we can anneal before the contamination exceeds our limit?
    c. Undo the recipe change of b). Another way to reduce diffusion is to increase gate oxide thickness. What is the minimum thickness of the gate oxide needed to keep metal diffusion below the limit? Make a graph to support your conclusion.
2. Please open Assignment4_GateLast.zam. Build the project. Open the viewer. Understand the structure. Look at the cross-section. Look at the process steps. Notice that the gate stack is deposited AFTER the high temperature anneal steps for the source and drain.
    a. The polysilicon dummy gate is refractory, so does not diffuse. Diffusion only occurs during subsequent high-temperature steps. We have added a dummy anneal at the end to represent the combined effect of the future steps. Plot the Al metal diffusion in the structures after the dummy anneal. What is the concentration in the active region?
    b. The device stops working if the metal contamination exceeds 10^{12} cm-3. Go to the dummy anneal recipe and increase the anneal temperature. What maximum temperature can we anneal before the contamination exceeds our limit?
    c. Undo the recipe change of b). The gate last process solves the problem but at the expense of added complexity. List down the various extra steps and materials we add.
    d. One of the critical requirements of the metal-last process is the high-aspect-ratio metal deposition. Change the length of the channel. At what point does the process fail?
3. Open Inbuilt example of 14nmFinFET.zam.
    a. Can you identify the steps where the gate is deposited?
    b. Is it a gate first or gate last process?
    c. What is the gate-metal material for PMOS and NMOS? Note that it is different.